

DAHOUN Manel

Computer Systems engineering

BCI brain computer interface, Software developer and UI/UX designer with computer systems engineering ‘s degree studies at Hight school of computer science 08 may 1945 Sidi Bel Abbes and hands-on work across web, mobile and desktop applications. Combines frontend implementation with product-interface design, teamwork and research in EEG-based biometric authentication.

Personal info

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Languages

- Arabic
- French
- English

Education

Nov 2024 – Jun 2026

ESI-SBA

École Supérieure en Informatique 08 Mai 1945 Sidi Bel Abbès · Algeria

Computer science engineering studies.
got both engineering and master’s degree in Computer systems engineering

Experience

Nov 2024 – Jun 2026

INGENIUMS

Technical Team Member, Developer & UI/UX Designer

Designed UI and coded selected interfaces for IngeCode 1v1 and the IngeHack registration website; mentored IngeCode participants on challenges and coding topics.

JavaScript · Next.js · Tailwind CSS · Figma

2025 – 2026

INGEHACK 4.0 / 5.0

Technical & Event Roles

4.0: designed pages and visual assets, coded website sections and supported event-day execution. 5.0: managed event-day execution as part of the Technical Team.

Since Feb 2025

ESI-SBA FABLAB

Robotics Laboratory Member

Took part in embedded-systems and robotics projects, lab work and presentations.

ESP32 · Raspberry Pi · Pixhawk · Electronics · CAD

2023 · 2024 · 2025

LINUX INSTALL PARTY

Technical Support

Completed 20+ installations across editions: single-boot, dual-boot and virtual-machine setups, plus troubleshooting.

Skills

Programming

C, Python, JavaScript, C++, Java, OCaml, Pascal

Web & mobile

React, Next.js, HTML, CSS, Tailwind CSS, Flutter

Backend & data

Node.js, Express, Fastify, Prisma, PostgreSQL, SQLite, MongoDB, MySQL, Cassandra, Neo4j

Desktop & workflow

Electron, Linux, Git, GitHub, SSH, Makefiles, virtual machines

AI & research

OpenCV, ArUco, AI agents, MNE, PsychoPy, pyRiemann, Emotiv Insight 2

Embedded

ESP32, Raspberry Pi, Pixhawk, Arduino, electronics, CAD, sensors

Design

Figma, Photoshop, Illustrator, Canva

Networking

Intergiciel

Multi services, distributed systems

Projects

Jan – Jul 2025

EEG BIOMETRIC AUTHENTICATION

ESI-SBA / FabLab · Research prototype

Built a prototype for biometric verification using a low-channel wearable EEG headset and personalized visual stimuli, studying P300 responses and Riemannian representations. Covered literature review, data collection, experiments and analysis.

Python · Emotiv Insight 2 / Cortex · MNE · PsychoPy · pyRiemann . Data Collection

Sep 2025

DESKTOP APPLICATION PORTFOLIO

Freelance · Client confidential

Designed and developed four desktop applications: two multi-user apps sharing a database over a local network and two local apps.

Electron . React . Nodejs . Express . Festify . Prisma . PostgreSql . Centralisation . LAN

Mar – Jun 2025

INTERSTELLAR TEAM — DRONE VISION

Team of 5

Implemented and demonstrated a computer-vision line-detection and line-following component.

Computer vision · ArUco

Nov 2025

DEPANINI

On-demand roadside assistance app

Designed the product experience and built the frontend for customer, administrator and service-provider / driver interfaces.

Flutter · Figma

Mar – Jun 2024

LES ŒUVRES SOCIALES

Social services platform · Team of 5

Handled UI/UX design and frontend development for a platform centralizing social-service administration and employee offers; connected the frontend to the backend.

React · Figma

2024 – 2026

INGECODE 1V1 & INGEHACK PLATFORMS

Ingeniums Technical Team · Team of 5

Designed platform screens and coded frontend interfaces for the coding challenge and hackathon registration experience.

JavaScript · Next.js · Tailwind CSS · Figma

2023

PROGRAMMING LANGUAGE DOCS WEBSITE

Team of 5

Delivered the complete frontend, visual design, logo and visual identity, plus a Bash testing script.

HTML · CSS · JavaScript · Bash

Recognition

1st Place

Mobadarat Jisr Numuw · Jul 2025

2nd Place

KODIKOS Hackathon · Nov 2025

Finalist

Northian Crows (cryptography & algorithms) · Mar 2023

AFROBOT

Fighting robot “Wall-E” · Jul 2025 — electronics, finance and component purchasing

AIQuest’25 / Alpha AI

QR-transaction fraud detection; AI-agent and computer-vision challenges · 2025

Member of the Month

Ingeniums · Oct 2024

Internships

15 – 30 Sep 2024

ASAL — ALGERIAN SPACE AGENCY

SAC Department · Ouled Fayet

Developed a desktop application and gained exposure to satellite maintenance and satellite-utilization management.

12 – 20 Jul 2024

THINKTHEC — KANOUNGPT

Development internship · UI/UX user interface/ user experience

Designed the interface for a legal chatbot answering questions about selected Algerian laws.

Figma

Teaching & Leadership

Since Sep 2025

ONLINE INSTRUCTOR

C Programming & Algorithmics

Teach pointers, arrays, file handling, processes, dynamic memory and Makefiles in C; algorithmics and complexity with Python, including trees and graphs.

C · Python

Sep 2025 – Jun 2026

INGENIUMS — LEARNING COMMUNITY

Vice Leader

Helped recruit tutors, plan workshops, prepare venues and coordinate learning activities.

26 Aug – 18 Sep 2025

ETHICAL HACKING FUNDAMENTALS

Online Tutor

Individual instruction in networking, Linux, web security, practical labs, algorithmics and DNS.

2 Dec 2024

GIT & GITHUB WORKSHOP

Tutor

Delivered an introductory workshop on Git/GitHub fundamentals, SSH-key setup and common commands, with a practical mini-lab.